AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 4)

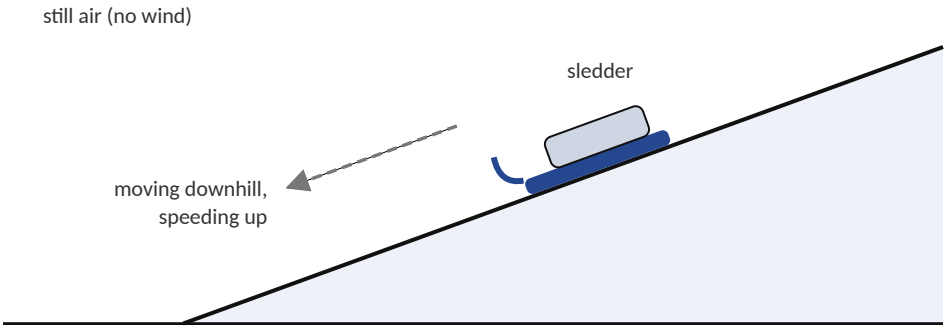
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

For all questions:

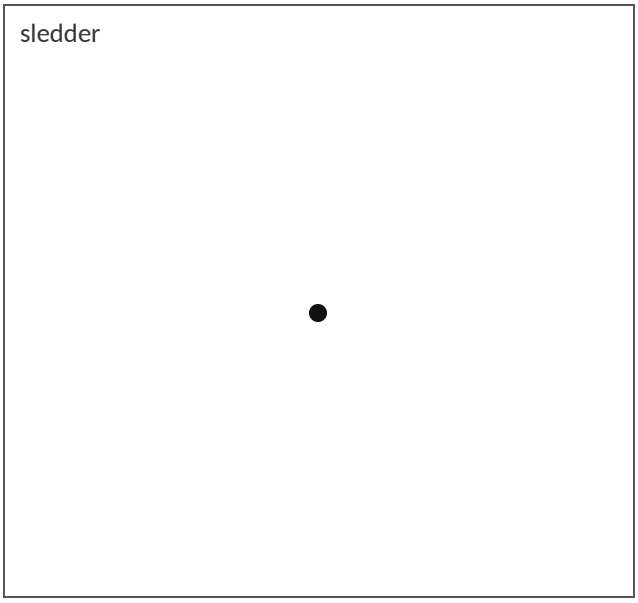
- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 25 points each (100 points total).

1. (25 points) A kid on a cardboard sled slides down a grassy hill. In the picture the sledder moves down the slope to the left and is speeding up. Friction from the grass and drag from the air are **not** negligible. No angle is given.



Assumptions for this practice problem: *the air is still (no wind), so the air moves past the sledder opposite its motion.*

Draw and label the free-body diagram of the sledder. Draw the forces themselves, not components. Each force is one arrow that starts on the dot and points in the direction the force acts. No calculations.



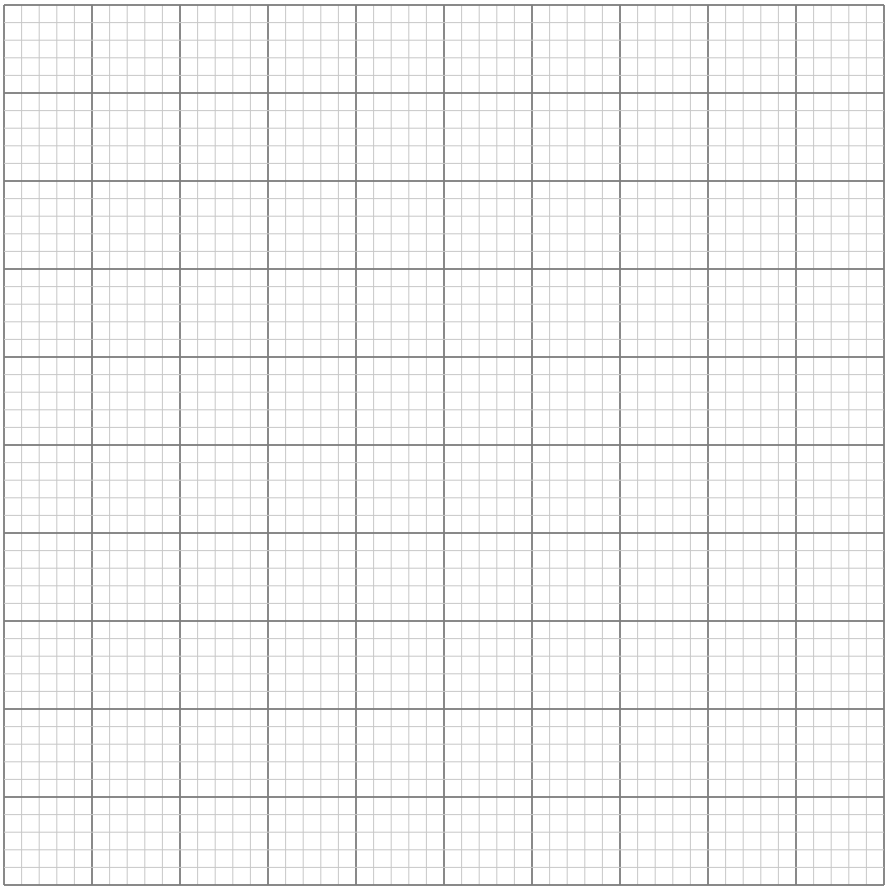
2. (25 points) Students measure the drag force F_D on a speed skater in a wind tunnel at several air speeds v. The drag force is modeled by $F_D = Kv^2$, where K is a constant.

Speed v (m/s)	Drag force F_D (N)	
4.00	2.89	
6.00	5.76	
8.00	10.4	
10.0	17.4	
12.0	24.1	
14.0	34.0	

(a) Using $F_D = Kv^2$, state the quantity you will plot on the vertical axis and the quantity you will plot on the horizontal axis so the data form a straight line, each with its units. State what the slope equals.

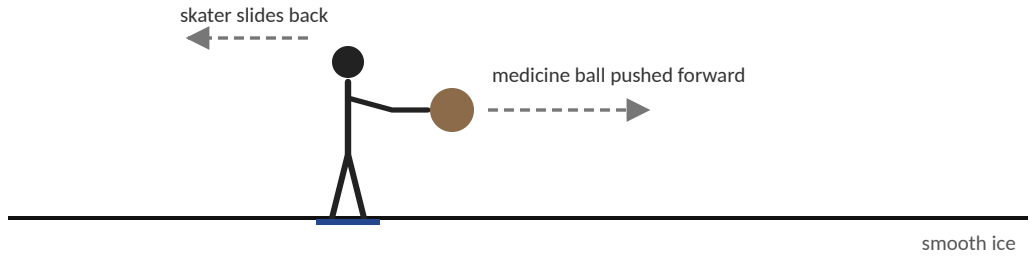
(b) Use the blank column in the table to calculate the values you will plot on the horizontal axis.

(c) Plot the data on the grid. Label each axis with the quantity and units, choose a scale, and draw the best-fit line.



(d) Calculate the slope of your best-fit line using two well-separated points on the line. Then give K with its units.

3. (25 points) A skater standing on smooth ice pushes a medicine ball horizontally to the right. Find the skater's acceleration during the push.



medicine ball mass	5.00 kg
skater mass	55.0 kg
distance her hands push the ball, relative to her body	0.600 m
speed the ball leaves her hands, relative to her	3.00 m/s

Both the push distance and the release speed are measured relative to the skater, who is herself sliding backward.

Assumptions for this practice problem: *the skater starts at rest and slides freely on the ice (ignore friction and air resistance); the medicine ball starts from rest; its acceleration during the push is constant; the push between the medicine ball and the skater is the only horizontal interaction.*

(a) Draw the free-body diagram of the skater during the push.

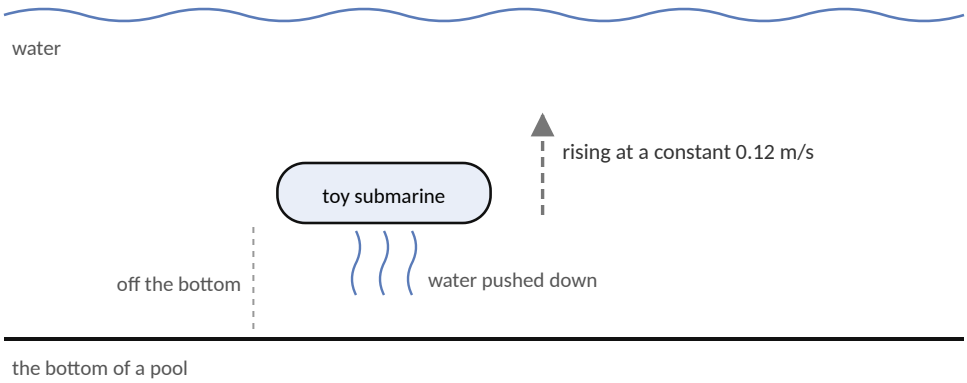
(a) skater

(b) Calculate the medicine ball's acceleration relative to the skater.

(c) Relative to the skater, the medicine ball's acceleration is the sum of the magnitudes of the medicine ball's and the skater's accelerations relative to the ground. Use Newton's second and third laws to find the medicine ball's acceleration relative to the ground, and the force the skater exerts on it.

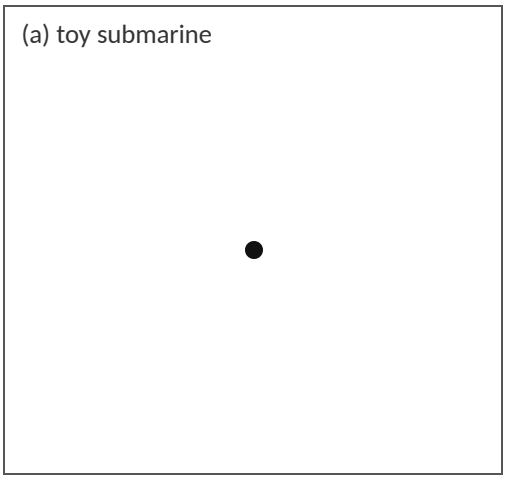
(d) Calculate the skater's acceleration, magnitude and direction.

4. (25 points) A 0.800 kg toy submarine has lifted off the bottom of a pool. Its propeller pushes water downward, so the water pushes it upward with a thrust of 1.40 N. After it leaves the bottom, it rises at a constant speed of 0.12 m/s.



Assumptions for this practice problem: *water drag on the toy submarine is negligible; $g = 9.8\ m/s^2$.*

(a) Draw the free-body diagram of the toy submarine after it has left the bottom.



(b) Find the toy submarine's acceleration.

(c) With up as positive, write Newton's second law for the toy submarine and solve it for the buoyant force in letters.

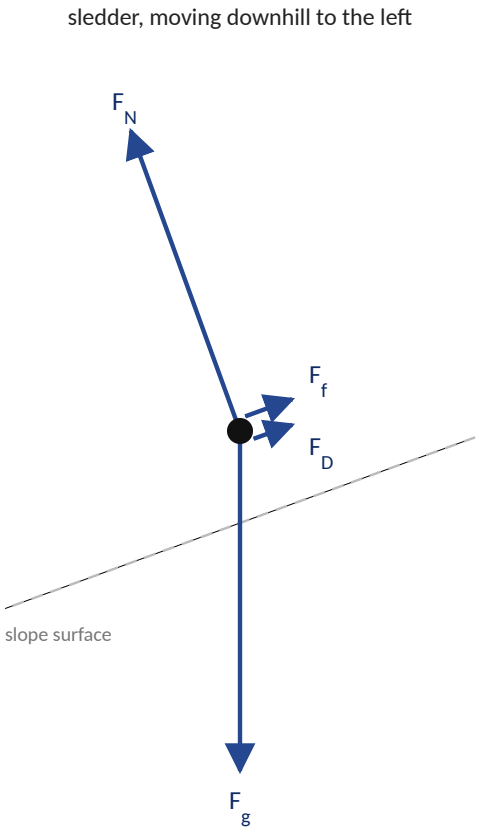
(d) Calculate the magnitude of the buoyant force on the toy submarine.

Answer Key – Practice Test, Version 4 – Introductory Dynamics

Worked step by step: diagram, symbolic equations and algebra first, then numbers with units. $g = 9.8\ m/s^2$ does not limit significant figures; answers match the significant figures of the givens.

Numerical problems: diagram first, then a = ΣF/m (equation sheet), written $a = F_{net}/m$. Replace F_{net} with the actual forces and their signs, swap $F_g = mg$, solve in letters, then plug in numbers with units.
Kinematics: #1 $v = v_0 + at$ · #2 $\Delta x = v_0 t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Free-body diagram on a slope (25 points)



Step 1 **List every force on the sledder: exactly four.**

Earth pulls on it: F_g. The grass pushes on it two ways: F_N (perpendicular to the surface) and kinetic friction F_f (along the surface). The air pushes on it: drag F_D.

Step 2 **Directions. No angle is needed.**

F_g: straight down, vertical, not perpendicular to the slope.

F_N: perpendicular to the slope, pointing out of the surface, so it tilts away from vertical toward the left.

F_f: along the slope, uphill (up and to the right), opposite the sliding.

F_D: opposite the motion through the air. The air is still and the sledder moves downhill, so drag also points uphill, parallel to F_f.

Step 3 **Check the diagram.**

Four arrows, each starting on the dot, each labeled. F_f and F_D are two separate arrows, side by side, both uphill. Speeding up, slowing down or moving at a constant speed changes the arrow lengths, not their directions.

Lengths in this key: speeding up downhill, so the two uphill arrows together are shorter than the part of gravity that pulls along the slope, and FN is a little shorter than Fg. Lengths are not graded.

Watch out: do not add the components of F_g (along and perpendicular to the slope) as extra arrows; they are not new forces. Drag is the air resistance, not a fifth force, and there is no forward "motion" force.

2. Drag linearization (25 points)

Step 1 **(a) Match $F_D = Kv^2$ to $y = mx + b$.**

$F_D = K \cdot (v^2) + 0$

vertical (y)	horizontal (x)	slope	intercept
F_D (N)	v^2 (m^2/s^2)	K	0

Vertical: F_D in N. Horizontal: v^2 in m^2/s^2. Slope = K.

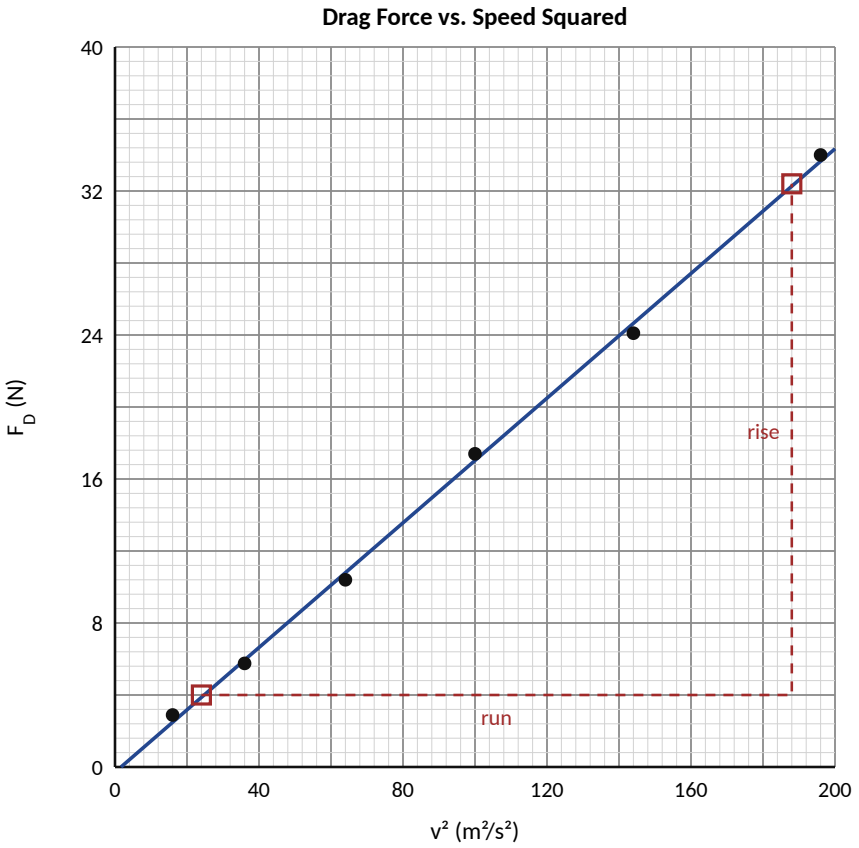
Watch out: v^2 is speed squared, in m^2/s^2. It is not an acceleration (m/s^2).

Step 2 **(b) Square each speed; the units square too.**

Example: $(4.00\ m/s)^2 = 16.0\ m^2/s^2$

v (m/s)	F_D (N)	v^2 (m^2/s^2)
4.00	2.89	16.0
6.00	5.76	36.0
8.00	10.4	64.0
10.0	17.4	100
12.0	24.1	144
14.0	34.0	196

Step 3 **(c) Graph: labeled axes with units, a scale that uses more than half the grid, one straight line through the middle of the points.**



The best-fit line is not forced through any data point. The red squares are points read off the line, not data points.

Step 4 **(d) Slope from two well-separated points on the line, units carried.**

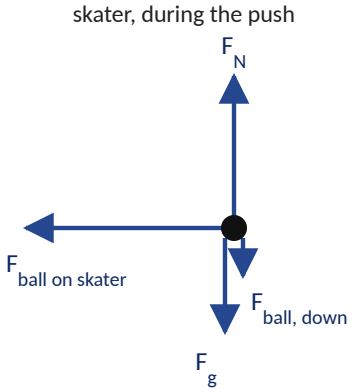
Points read off the line, away from data points: (24 m²/s², 4.0 N) and (188 m²/s², 32.4 N)

$$\text{slope} = \frac{32.4\ \text{N} - 4.0\ \text{N}}{188\ \text{m}^2/\text{s}^2 - 24\ \text{m}^2/\text{s}^2} = \frac{28.4\ \text{N}}{164\ \text{m}^2/\text{s}^2} = 0.17317\ \text{N}\cdot\text{s}^2/\text{m}^2$$

$K \approx 0.173\ \text{N}\cdot\text{s}^2/\text{m}^2\ (= \text{kg/m})$. Any value from 0.159 to 0.187 N·s²/m² from a reasonable line earns credit.

Units: N ÷ (m²/s²) = N·s²/m², and N = kg·m/s², so N·s²/m² = kg/m.

3. Recoil (25 points)



Step 1 **(a) Forces on the skater during the push.**

Vertical: the ice pushes up (F_N); Earth pulls down (F_g); the medicine ball presses down ($F_{ball,\ down}$). Nothing accelerates vertically, so $F_N = F_g + F_{ball,\ down}$.

Horizontal: the only force is the medicine ball pushing back on the skater, $F_{ball\ on\ skater}$, to the left. The vertical forces do not enter the horizontal net-force equation. Arrow lengths are not to scale.

Step 2 **Units and frames.**

All values are already in SI units.

Step 3 **(b) Kinematics in the skater's frame.**

Relative to the skater: $v_0 = 0$, $v_{rel} = 3.00$ m/s over $L = 0.600$ m. Use #3 in her frame.

$$a_{rel} = \frac{v_{rel}^2}{2L} = \frac{(3.00\ \text{m/s})^2}{2(0.600\ \text{m})} = \frac{9.00\ \text{m}^2/\text{s}^2}{1.20\ \text{m}} = \boxed{7.50\ \text{m/s}^2}$$

Step 4 **(c) Second and third laws, letters first.**

Third law: the ball and the skater push on each other with forces of equal size F. Second law on each: $F = m_{ball}a_{ball}$ and $F = M\,a_{skater}$, so $m_{ball}a_{ball} = M\,a_{skater}$.

They accelerate in opposite directions, so relative to the skater: $a_{rel} = a_{ball} + a_{skater} = a_{ball} + \frac{m_{ball}a_{ball}}{M}$

Solve for the ball: $a_{ball} = a_{rel}\frac{M}{M + m_{ball}}$

$$a_{ball} = (7.50\ \text{m/s}^2)\frac{55.0\ \text{kg}}{55.0\ \text{kg} + 5.00\ \text{kg}} = 6.875\ \text{m/s}^2 \rightarrow \boxed{6.88\ \text{m/s}^2}$$ relative to the ground

$$F = m_{ball}a_{ball} = (5.00\ \text{kg})(6.875\ \text{m/s}^2) = 34.375\ \text{N} \rightarrow \boxed{34.4\ \text{N}}$$

Step 5 **(d) Second law on the skater.**

$$a_{skater} = \frac{F}{M} = \frac{34.375\ \text{N}}{55.0\ \text{kg}} = \boxed{0.625\ \text{m/s}^2\text{, to the left (backward)}}$$

Check: $6.875\ \text{m/s}^2 + 0.625\ \text{m/s}^2 = 7.50\ \text{m/s}^2 = a_{rel}$.

Watch out: the 0.600 m and 3.00 m/s are measured relative to the skater. Treating them as ground values gives 7.50 m/s² for the ball, about 9% too big.

4. Thrust and buoyancy (25 points)

toy submarine, off the bottom



Step 1 **(a) Forces on the toy submarine after it leaves the bottom.**

F_g down (Earth). Thrust F_t up (the water it pushes down pushes back up on it). Buoyancy F_b up (the surrounding water).

No normal force: it is no longer touching the bottom. Constant velocity means a = 0, so $F_t + F_b$ together equal F_g.

Step 2 **(b) Acceleration from the motion data.**

Constant speed in a straight line means the velocity is not changing: **a = 0**

Here a = 0 comes from the stated constant speed, not from the fact that it moves upward.

Step 3 **(c) Newton's second law, up is +, letters first.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_b + F_t - F_g}{m} \rightarrow F_b + F_t - mg = ma$$

$F_b = m(g + a) - F_t$

Step 4 **(d) Now the numbers, units in every parenthesis.**

$F_b = (0.800\ kg)(9.8\ m/s^2 + (0\ m/s^2)) - 1.40\ N = (0.800\ kg)(9.8\ m/s^2) - 1.40\ N = 7.840\ N - 1.40\ N = 6.440\ N$

$F_b = 6.44$ N (the subtraction sets the decimal place; g does not limit the significant figures).

Watch out: a = 0 here only because the speed is constant, not because the toy submarine moves up. Dropping the thrust gives the wrong buoyant force.

1. Free-body diagram on a slope

Suggested partial credit	pts
F_g straight down (vertical), labeled	5
F_N perpendicular to the slope, out of the surface, labeled	5
F_f along the slope, uphill, labeled	5
F_D uphill (opposite the motion through still air), labeled, separate from F_f	5
Every arrow starts on the dot; no components and no extra forces	5
Total	**25**

2. Drag linearization

Suggested partial credit	pts
(a) F_D (N) on the vertical axis, v^2 (m^2/s^2) on the horizontal axis	4
(a) Slope = K	2
(b) v^2 column correct, with units	4
(c) Axes labeled with quantity and units	2
(c) Scale uses more than half the grid	2
(c) Points plotted correctly	2
(c) One straight best-fit line	2
(d) Two well-separated points read off the line, not data points	3
(d) K from 0.159 to 0.187, with units $N \cdot s^2/m^2$ (or kg/m)	4
Total	**25**

3. Recoil

Suggested partial credit	pts
(a) F_N up, F_g down (the ball pressing down may also be shown); ball-on-skater force backward; no forward force	5
(b) $a_{rel} = v_{rel}^2/(2L) = 7.50\ m/s^2$	5
(c) $m_{ball}a_{ball} = M\ a_{skater}$ from the second and third laws	5
(c) $a_{ball} = 6.88\ m/s^2$ and F = 34.4 N	4
(d) $a_{skater} = 0.625\ m/s^2$, backward	6
Total	**25**

4. Thrust and buoyancy

Suggested partial credit	pts
(a) F_g down, thrust and buoyancy up, labeled	4
(a) No normal force	2
Acceleration: a = 0 from the constant speed	6
$F_b + F_t - mg = ma$ written with up as +	4
Solved in letters: $F_b = m(g + a) - F_t$	3
Substitution with units in every parenthesis (and grams to kg if needed)	3
$F_b = 6.44$ N	3
Total	**25**